Reproducing DistilBERT: Verifying Efficiency-Performance Tradeoffs on IMDb Sentiment Classification

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1. Introduction

Large pretrained language models such as BERT have become central to modern natural language processing because they achieve strong performance across many downstream tasks. However, these models are expensive to train, store, and deploy. BERT-base contains roughly 110 million parameters, making inference slower and more computationally expensive, especially in settings where latency, memory usage, or deployment cost matters.

Our project reproduces key results from *DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter* by Sanh, Debut, Chaumond, and Wolf. The paper proposes DistilBERT, a compressed version of BERT trained using knowledge distillation. In this setup, a smaller student model learns to imitate the behavior of a larger teacher model. DistilBERT reduces BERT-base from 12 Transformer layers to 6, removes token-type embeddings and the pooler, and initializes the student model from every other layer of BERT-base. During pretraining, it uses a triple loss objective combining distillation loss, masked language modeling loss, and cosine embedding loss.

The main claim of the paper is that DistilBERT preserves most of BERT's language understanding ability while being significantly smaller and faster. Specifically, the paper reports that DistilBERT retains approximately 97% of BERT's performance while being 40% smaller and 60% faster. Our goal was to test whether this efficiency-performance tradeoff holds in a controlled downstream sentiment classification setting.

2. Chosen Result

We chose to reproduce the IMDb binary sentiment classification result from the DistilBERT paper, along with the model size and inference speed comparisons. This corresponds to the paper's reported downstream task accuracy in Table 2 and efficiency comparison in Table 3.

The original paper reports that DistilBERT achieves 92.82% test accuracy on IMDb, compared to 93.46% for BERT-base. This is only a 0.64 percentage point gap. The paper also reports that DistilBERT has 66 million parameters compared to 110 million for BERT-base, making it approximately 40% smaller. For CPU inference, the paper reports 410 seconds for DistilBERT compared to 668 seconds for BERT-base.

We chose IMDb because it directly tests the paper's central claim: whether a smaller distilled model can preserve the downstream performance of a larger teacher model. IMDb is also practical for reproduction because it is a standard binary sentiment classification dataset and does not require reproducing the full GLUE benchmark or pretraining DistilBERT from scratch.

3. Methodology

We fine-tuned two pretrained models: distilbert-base-uncased and bert-base-uncased, both from HuggingFace Transformers. We used the full IMDb dataset, which contains 25,000 training examples and 25,000 test examples with balanced positive and negative sentiment labels. The dataset was accessed through HuggingFace Datasets.

We did not reproduce DistilBERT pretraining from scratch because the original paper required 8 V100 GPUs for approximately 90 hours. Instead, we used the publicly released pretrained DistilBERT weights, which is consistent with the intended downstream use of the model.

Both models were fine-tuned under identical conditions. We trained for 3 epochs with a learning rate of 2e-5, batch size of 32, maximum sequence length of 512, weight decay of 0.01, warmup ratio of 0.1, and fp16 mixed precision on CUDA. Training was done sequentially using the HuggingFace Trainer API, and accuracy was computed using the evaluate library.

For evaluation, we measured test accuracy on the full IMDb test set. We also measured parameter count, FP32 model size, GPU inference latency, CPU inference latency, and training time. GPU latency was measured using 200 forward passes with batch size 1 after 20 warmup passes. CPU latency was measured using 50 forward passes with batch size 1. Unlike the original paper, which only reports CPU inference time, our evaluation includes both CPU and GPU inference to better understand whether DistilBERT's speedup generalizes across hardware.

4. Results and Analysis

Our results support the main claims of the DistilBERT paper. DistilBERT retained nearly all of BERT-base's IMDb accuracy while being much smaller and faster.

Metric: IMDb Test Accuracy

DistilBERT: 93.06%

BERT-base: 94.11%

Our Result: 98.9% retained

Paper Claim: 97% retained

Metric: Parameters

DistilBERT: 67.0M

BERT-base: 109.5M

Our Result: 38.8% fewer

Paper Claim: 40% fewer

Metric: Model Size, FP32

DistilBERT: 255 MB

BERT-base: 418 MB

Our Result: 38.8% smaller

Paper Claim: Not reported

Metric: GPU Inference Latency

DistilBERT: 6.58 ms

BERT-base: 12.46 ms

Our Result: 1.89x faster

Paper Claim: Not reported

Metric: CPU Inference Latency

DistilBERT: 481.9 ms

BERT-base: 953.5 ms

Our Result: 1.98x faster

Paper Claim: 1.63x faster

Metric: Training Time

DistilBERT: 19.5 min

BERT-base: 38.6 min

Our Result: 1.98x faster

Paper Claim: Not reported

DistilBERT achieved 93.06% test accuracy compared to 94.11% for BERT-base. This means DistilBERT retained 98.9% of BERT-base's accuracy, which slightly exceeds the paper's reported 97% retention claim. The remaining accuracy gap was only 1.05 percentage points, showing that the smaller model preserved most of the larger model's downstream performance.

The parameter results also closely matched the original paper. We measured 67.0 million parameters for DistilBERT and 109.5 million for BERT-base, corresponding to a 38.8% reduction. This is very close to the paper's 40% smaller claim. The small discrepancy is likely due to rounding and differences between reported model sizes and current HuggingFace model implementations.

The speed results were even stronger than expected. On GPU, DistilBERT was 1.89x faster than BERT-base. On CPU, it was 1.98x faster. The CPU speedup exceeded the paper's reported 1.63x speedup. This difference likely comes from hardware differences and improvements in modern PyTorch and HuggingFace implementations since the paper was released in 2019.

The fact that GPU and CPU speedups were similar is important. It suggests that DistilBERT's efficiency advantage comes primarily from architectural simplification, especially using 6 Transformer layers instead of 12, rather than from hardware-specific effects. In other words, DistilBERT is faster because it performs less computation while still preserving most of BERT's learned language representations.

The main challenge was compute time. Even with a modern GPU, fine-tuning both models on the full IMDb dataset took almost one hour. BERT-base took about 38.6 minutes, which limited how much hyperparameter tuning we could realistically perform. We also observed expected warnings about missing or unexpected keys when loading pretrained models with new classification heads. These warnings are normal during fine-tuning and did not indicate a problem with the experiment.

5. Reflections

This project showed that knowledge distillation is a practical and effective strategy for model compression. Before running the experiment, we expected DistilBERT to perform slightly worse than the paper's reported result because of implementation differences and changes in training tools over time. Instead, DistilBERT retained 98.9% of BERT-base accuracy, which was stronger than expected.

The most important takeaway is that efficiency gains do not always require a large sacrifice in accuracy. DistilBERT was almost twice as fast as BERT-base while losing only about one percentage point of accuracy on IMDb. This makes distilled models especially useful for real-world deployment, where latency, memory usage, and cost often matter as much as raw performance.

A second takeaway is that benchmarking across hardware matters. Since the original paper focused on CPU inference, our additional GPU measurements helped show that the speedup is not limited to one environment. DistilBERT remained faster on both CPU and GPU, strengthening the argument that its advantage comes from the model design itself.

Future work could extend this reproduction in several directions. First, task-specific distillation during fine-tuning could potentially recover the remaining accuracy gap. Second, evaluating on additional GLUE tasks such as SST-2 or MRPC would show whether the IMDb results generalize to other language understanding tasks. Third, post-training quantization could reduce memory usage and inference cost even further. Finally, applying the same distillation approach to newer encoder models such as RoBERTa would test whether the benefits of knowledge distillation generalize beyond BERT.

6. References

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Github:

https://github.com/EcSky19/DistilBERT-Replication